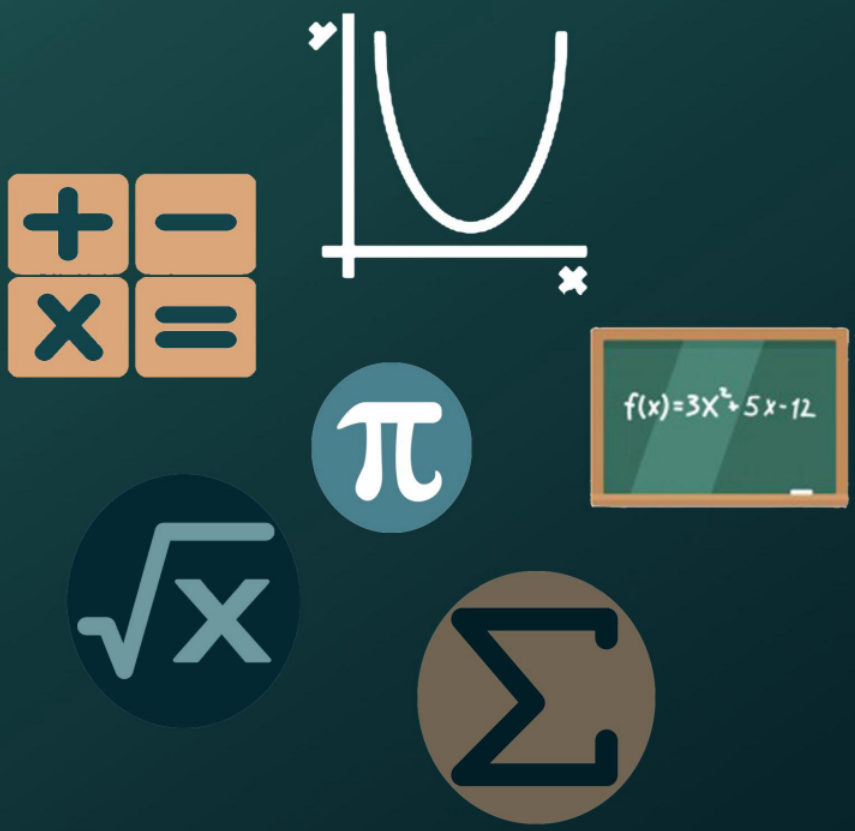
MATHEMATICS

# Principle of Mathematical Induction




SHEET

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# *Contents*


| | | |
|---|---|---|
| 1. | Module Description | Page - ii |
| 2. | Homework Index | Page - iii |
| 3. | Exercise 1 | Page - 1 |
| 4. | Exercise 2 | Page - 2 |


***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 [redacted] Contains Single Correct Type questions

1A [redacted] Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 [redacted] Single Correct Type questions.

2A [redacted] Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM [redacted] JEE Mains previous years topic wise questions.

JA [redacted] JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| Sr. No. | Topics | Ex-1 | Ex-2 |
|---|---|---|---|
| 1 | Subjective Approach to induction | 1-21 | 1-4 |
| 2 | Objective Apporoach to induction | | |

## Exercise - 1 — Concept Builder

**SUBJECTIVE TYPE**

**Q.1** If P(n) stands for the statement "$n(n+1)(n+2)$ is divisible by 6", then what is P (3) ?

**Q.2** Let P(n) be the statement "$n^2+n$ is an odd integer". Show that if P(m) is true, then $P(m+1)$ is also true.

**Q.3** Prove by the principle of mathematical induction that for all

$$n\in N: 1+4+7+\ldots+(3n-2)=\frac{1}{2}n(3n-1).$$

**Q.4.** Prove by induction that :

$$1.2+2.3+3.4+\ldots+n.(n+1)$$

$$=\left[\frac{n(n+1)(n+2)}{3}\right].$$

**Q.5** Prove by induction that :

$$1.2.3+2.3.4+\ldots+n(n+1)(n+2)$$

$$=\frac{n(n+1)(n+2)(n+3)}{4}.$$

**Q.6** Prove by induction that :

$$3\times6+6\times9+9\times12+\ldots+(3n)\times(3n+3)$$

$$=3n(n+1)(n+2).$$

**Q.7** Prove by induction that :

$$1+3+3^2+\ldots+3^{n-1}=\frac{(3^n-1)}{2}.$$

**Q.8** Prove by induction that :

$$1^3+2^3+3^3+\ldots+n^3=\left(\frac{n(n+1)}{2}\right)^2.$$

**Q.9** Prove by induction that :

$$1.3+2.3^2+3.3^3+\ldots+n.3^n=\frac{(2n-1)3^{n+1}+3}{4}.$$

**Q.10** Prove by induction that :

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots+\frac{1}{2^n}=1-\frac{1}{2^n}.$$

**Q.11** Prove by induction that :

$$a+ar+ar^2+\ldots+ar^{n-1}=\frac{a(r^n-1)}{r-1}.$$

**Q.12** Prove by induction that :

$$\left(1+\frac{3}{1}\right)\left(1+\frac{5}{4}\right)\left(1+\frac{7}{9}\right)\ldots\left(1+\frac{(2n+1)}{n^2}\right)=(n+1)^2.$$

**Q.13** Prove by induction that :

$$1^2+3^2+5^2+\ldots+(2n-1)^2=\frac{n(2n-1)(2n+1)}{3}.$$

**Q.14** Prove by induction that :

$$1+2+3+\ldots+n<\frac{1}{8}(2n+1)^2.$$

**Q.15** Prove by induction that :

$$\frac{1}{1.4}+\frac{1}{4.7}+\frac{1}{7.10}+\ldots+\frac{1}{(3n-2)(3n+1)}$$

$$=\frac{n}{(3n+1)}.$$

**Q.16** Prove by induction that :

$$\frac{1}{3.5}+\frac{1}{5.7}+\frac{1}{7.9}+\ldots+\frac{1}{(2n+1)(2n+3)}$$

$$=\frac{n}{3(2n+3)}.$$

**Q.17** Prove by induction that :

$$\frac{1}{1.2.3}+\frac{1}{2.3.4}+\frac{1}{3.4.5}+...+\frac{1}{n(n+1)(n+2)}$$

$$=\frac{n(n+3)}{4(n+1)(n+2)}.$$

**Q.18** Prove by induction that :

$$\left(1+\frac{1}{1}\right)\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)...\left(1+\frac{1}{n}\right)=(n+1).$$

**Q.19** Prove by induction that :

$10^{2n-1}+1$ is divisible by 11.

**Q.20** Prove by induction that :

$3^{2n+2}-8n-9$ is divisible by 8.

**Q.21** Prove by induction that :

$41^n-14^n$ is a multiple of 27.

## *Exercise - 2* *Brain Booster*

**SUBJECTIVE TYPE**

**Q.1** Prove by induction that

$1!+2(2)!+3(3)!+...+n(n)!=(n+1)!-1$

for all natural numbers n.

**Q.2** Prove by induction that $\frac{n^5}{5}+\frac{n^3}{3}+\frac{7n}{15}$ is a natural number for all $n\in N$.

**Q.3** Prove by the principle of mathematical induction that for all $n\in N$,

$\sin\theta+\sin 2\theta+\sin 3\theta+...+\sin n\theta$

$$=\frac{\sin\left(\frac{n+1}{2}\right)\theta\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}.$$

**Q.4** Hence, by principle of mathematical induction P(n) is true for all $n\in N$,

$$1+\frac{1}{(1+2)}+\frac{1}{(1+2+3)}+...+\frac{1}{(1+2+3+...+n)}$$

$$=\frac{2n}{(n+1)}.$$